Design and Implementation of an End-to-End Data Engineering Pipeline for HR Employee Attrition Analysis Using Pentaho and Power BI

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***Abstract* — This study focused on designing and implementing a data engineering pipeline to help analyze employee attrition using Pentaho Data Integration and Power BI. The developed Pipeline helped automate the process of data extraction, transformation, and loading (ETL), as well as data modeling using a star schema approach. The dataset contained various human resource information, such as employee demographic characteristics, compensation information: monthly income and daily rates, as well as employee behavioral factors such as job satisfaction, work environment satisfaction and work-life balance. The preprocessing and data validation stages were carried out to ensure the quality, accuracy and consistency of the data used in the analysis process. After that, the transformed data was visualized with a Power BI interactive dashboard to help companies identify attrition patterns based on key factors, such as overtime status, level of work engagement, and level of employee satisfaction. The implementation results showed that a structured data engineering pipeline could support easier data management and provide a basis for making data-based decisions in human resource management. Furthermore, the resulting dashboard could also provide insights that can help organizations identify factors related to attrition so that they can develop more effective employee retention strategies.**

Keywords— data engineering, data pipeline, employee attrition, pentaho data integration, power BI, star schema, human resource management (HRM).

I. Introduction

Almost all companies today depend on technology and data to remain competitive in the industry. Even though technology continues to develop, human resources are still the most important thing in helping an organization succeed. However, unfortunately, high levels of employee attrition or turnover are still a challenge for many companies. High turnover rates can also cause decreased productivity, a disruption of workforce stability, and increased operational costs due to the process of recruiting, training, and onboarding new employees, which must be carried out repeatedly [1][4].

In addition, companies usually have large amounts of employee data; however, information that has the potential to provide insight into the factors causing attrition is still not utilized optimally. Employee data is usually stored in various sources, maintain in different formats, and suffers from poor data quality which can hinder the analysis and decision-making process. Therefore, data management solutions that can integrate, clean, and transform data into consistent and trustworthy information sources are urgently needed [3][5].

The development of data processing technology shows that we need data pipelines that can handle everything that happens to data from getting it in storing it changing it to finally presenting it in a way. This approach helps organizations manage their data better making it more efficient organized and able to grow with their needs, which in turn makes their business insights and decisions based on data. Modern data storage systems also make it easier to connect data processing parts with the platforms that organizations use to analyze their data.

Based on these problems, this research was carried out by designing and implementing a data pipeline from start to finish to analyze employee attrition using Pentaho Data Integration and Power BI. The pipeline created also automates the processes of ingestion, transformation, validation, and data modeling so that the resulting data was ready to be used to analyze the company's needs. This research also utilized employee attrition datasets to show that structured and well-designed data engineering could turn raw data into valuable information for the company.

As a final result, this research produced an interactive Power BI dashboard designed to provide an overview and information about employee attrition patterns based on demographic factors, compensation, job characteristics, and the level of satisfaction of employees working at the company. This informative dashboard visualization can help teams working on human resource management identify factors related to attrition and develop more effective employee retention strategies [8].

II. Data Source And Data Generation

The data used for this research came from the HR Analytics Employee Attrition & Performance dataset, which is available on Kaggle in CSV format. This dataset consists of 1,470 rows containing employee data and attributes that describe the characteristics of human resources in a company. The dataset has 35 columns, which include demographic information, job characteristics, salary, and satisfaction and performance indicators of employees working at the company.

The attributes and informations in the dataset can be grouped into three main categories with the following details:

- Personal Data, this data consists of basic information about employees who work at the company, such as Age, Gender, Marital Status, Educational Field, Education, Monthly Income, Monthly Rate, Daily Rate, Hourly Rate, Percent Salary Hike, Stock Option Level, Over 18, Employee Number, and Employee Count.
- Employment History and Status, this data consists of information about the work of employees working at the company, such as Job Role, Department, Job Level, Business Travel, Distance From Home, Total Working Years, Num Companies Worked, Years at Company, Years in Current Role, Years Since Last Promotion, Years With Current Manager, Training Times Last Year, Overtime, and Standard Hours.
- Job Satisfaction, Work Environment, and Retention, this data also consists of behavioral indicators and perceptions of employees who work at the company, such as Environment Satisfaction, Job Involvement, Relationship Satisfaction, Work-Life Balance, Job Satisfaction, Performance Rating, and Attrition.

This dataset has a variety of information that is suitable for analyzing employee attrition, starting from demographic aspects, salaries, job characteristics, and the level of satisfaction of employees working in the company. This information helps the process of identifying patterns and factors that have the potential to be related to attrition, so that it can provide useful information and support decision-making in the field of human resource management.

Before the pipeline starts, the dataset is first analyzed using Exploratory Data Analysis (EDA) using Python. This aims to understand the characteristics of the data, identify the data distribution of each attribute, check for outliers and anomalies in the data, and evaluate the quality of the data to be used. These EDA results become the basis and make it easier to determine the strategy that will be used when cleaning data, selecting the required attributes, and designing the transformation process in Pentaho.

The distribution of education level by field can be seen in Fighe visualization for numerical attributes can be seen in Figure 2, while the visualization for categorical attributes can be seen in Figure 3. The visualization shows that the dataset has quite different variations in employee characteristics, starting from demographic aspects, compensation, and also job characteristics. Apart from that, several attributes also show an uneven distribution, and there are several attributes that need to be considered at the transformation and data validation stages. These insights can be used as a reference when designing the data processing flow in the pipeline.

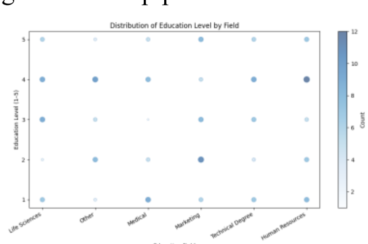

Fig. 1. Distribution of Education Level by Field

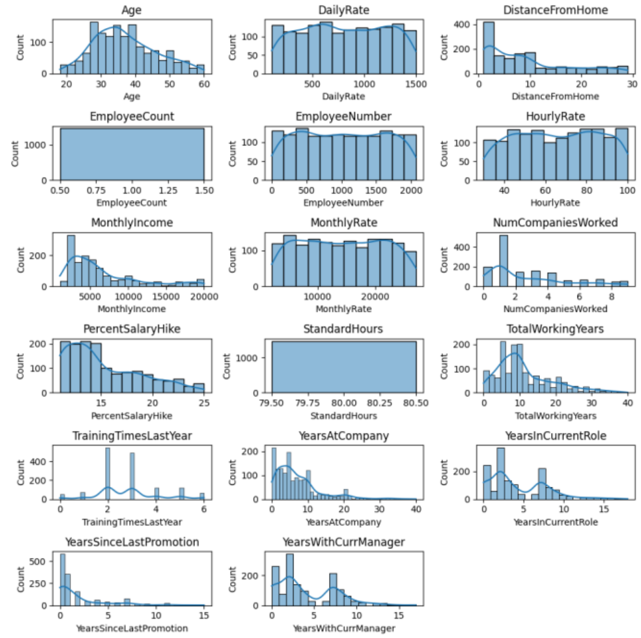

Fig. 2. EDA Visualization in Numerical Column

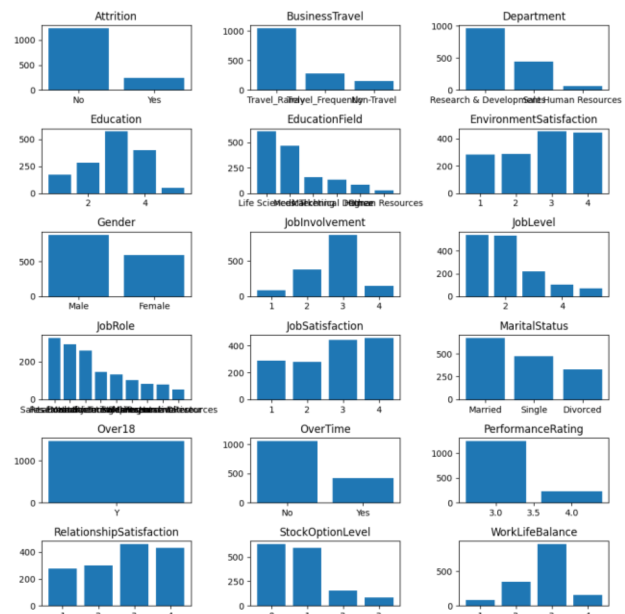

Fig. 3. EDA Visualization in Categorical Column

III. Data Engineering Pipeline Flow

A. Data Ingestion & Staging

The data ingestion and staging stages are the first processes carried out in this data engineering pipeline, aiming to move raw data from the data source to a new database environment that is ready to be used for analysis. In this study, the data source used is a Comma-Separated Values (CSV) file containing various kinds of company employee attrition information. All processes are carried out with Pentaho Data Integration (PDI) to ensure that data can be processed automatically, structured, consistent, and clean before being used for advanced analysis. The flow of the research process being developed can be seen in Figure 4.

First, the process begins by loading the raw data from the CSV file into the Online Transaction Processing (OLTP) database as a staging area. At the staging stage, after taking the input data, the system captures a load date to indicate which batch the data comes from and enters it into the staging data table in the OLTP database. After passing through the staging area, data from the staging table in OLTP is retrieved, and a data cleaning process is carried out to ensure and improve data quality before moving on to the next process. The implementation of the staging process in Pentaho can be seen in Figure 5.

At this cleaning stage, duplicate data is deleted. Apart from that, unnecessary attributes such as "Employee Count", "Standard Hours", and "Over 18" are removed because the employee count column only contains the number 1, the standard hours column contains the same number, namely 80 working hours, while the over 18 column contains "Yes" for all employee rows. A trimming process is also carried out to ensure that unnecessary characters and spaces are removed so that the data remains consistent. These things are done so that the data entering the system has a consistent format and there are no errors that could cause the analysis results to change. The implementation of the cleaning process in Pentaho can be seen in Figure 6.

After the cleaning process is complete, the validated data is moved from the OLTP database to the Online Analytical Processing (OLAP) database. At this stage, data modeling is carried out using a star schema approach to facilitate analysis and reporting. This star schema structure consists of a fact table which contains the main metrics and also several dimension tables which store descriptive information about the company's employees. Data is separated into fact tables and dimension tables to make the query process faster and make analysis between dimensions easier during the visualization process.

To help manage the data in the dimension tables, Slowly Changing Dimensions (SCD) and a lookup/update mechanism are implemented so that the system can add new data or update changed data automatically. Meanwhile, the fact table uses a full load strategy, namely reloading all data resulting from the transformation process every time the pipeline is run. This is done because the size of the dataset used is relatively small, so that the reloading process can still be carried out efficiently without placing a heavy burden on the system. In addition to helping keep data consistent, this approach also simplifies the synchronization process between the operational and OLAP environments.

Finally, the data serving process is carried out by integrating the OLAP database with Power BI. Data that has been arranged in a star schema form is used as the main data source to create analytical dashboards. Structured data helps the visualization process become more efficient and facilitates the process of creating interactive reports that can display various attrition patterns based on employee characteristics, level of job satisfaction, salary, and other factors relevant to the analysis.

With these data ingestion and staging stages, raw data that was previously scattered and unstructured can be converted into data that is consistent, integrated, and ready to be analyzed. In this way, the analysis process carried out in Power BI can produce more accurate information and support data-driven decision-making for managing human resources.

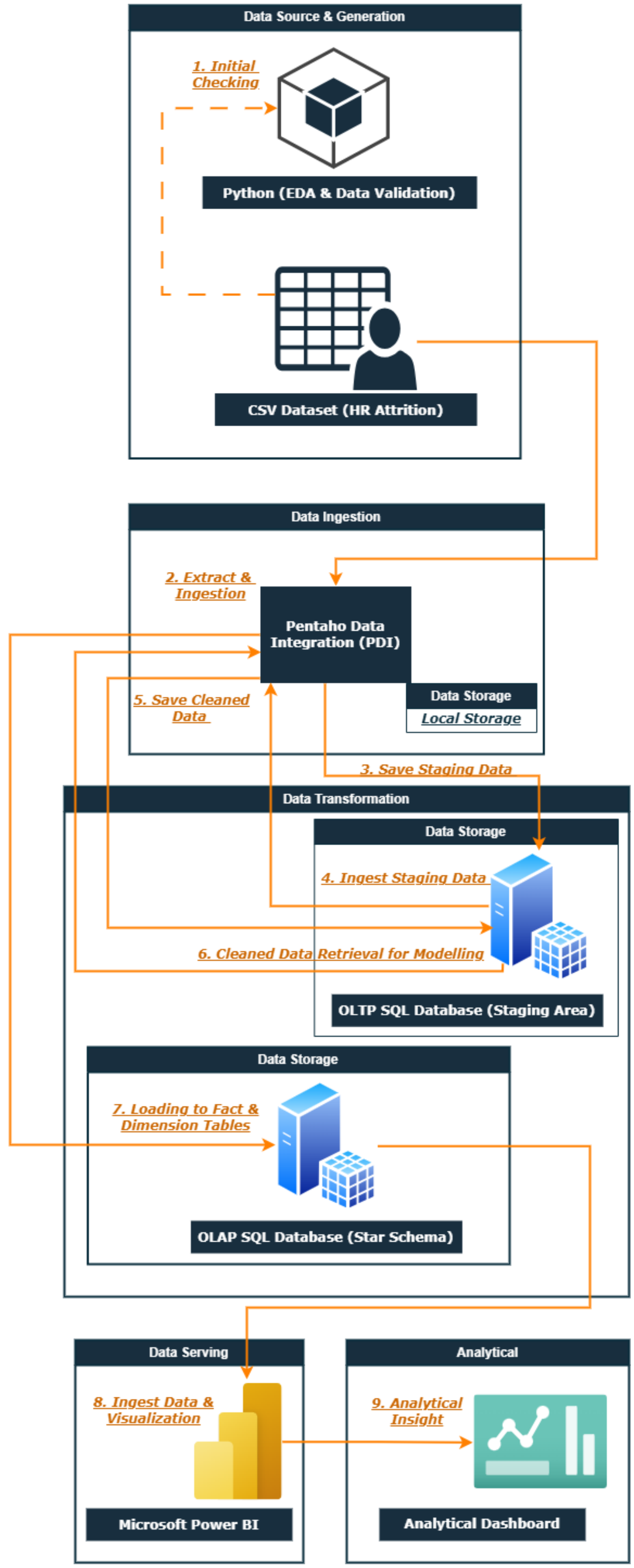

Fig. 4. Flowchart Research Framework

Fig. 5. Staging Area

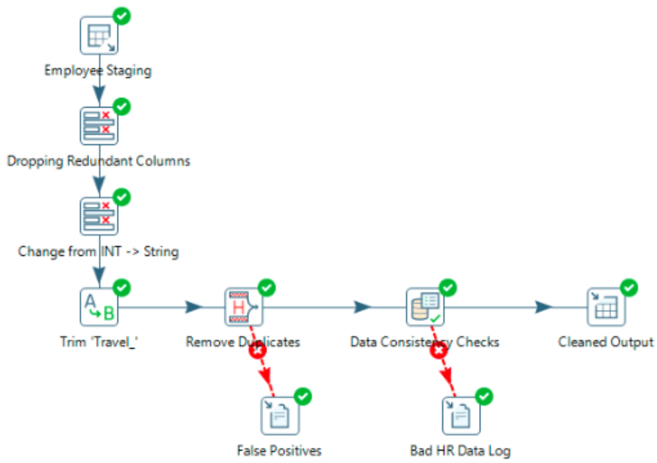

Fig. 6. Cleaning Visualization

B. Data Storage

The data storage design in this study was designed using a layered architecture. This was chosen so that the data could be managed in a structured manner, was easy to search, and could support analysis needs more efficiently. Data storage is divided into three main layers, landing zone, OLTP database as staging and cleaning area, and OLAP database for analytical.

The first layer is raw data storage also referred to as a landing zone. This layer's function is to store raw data in its original format. In this study, the HR Attrition dataset was saved in CSV format without prior modification. This aims to maintain the integrity of the original data and as a backup if the data process needs to be repeated from the beginning.

The second layer is the OLTP database which functions as a staging area. After the data is extracted from the CSV file, it is loaded into the OLTP database for temporary storage before entering the cleaning stage. During the cleaning stage, the clean data is also stored in the OLTP database before entering the modeling stage. The staging and cleaning areas allow data to be processed separately from the source data so that the risk of damaging the original data can be avoided. In addition, this separation helps the process of monitoring and controlling data while the pipeline is running.

The final layer is the OLAP database. This layer is used as a data repository for analysis and reporting needs. Data that has gone through the transformation and modeling process is loaded into a star schema-based data warehouse structure consisting of dimension tables and a fact table. This design was chosen because it can support multidimensional analysis processes and speed up query performance in applications.

Data storage separated into several layers was chosen because it provides several benefits, for example easier to track data, makes system maintenance easier, and reduces the computational load at the visualization stage. Because the staging, cleaning, transformation, and modeling processes have been completed before the data enters the OLAP database, Power BI can access structured data so that the analysis process becomes faster.

C. Data Transformation

After the data goes through the ingestion, staging, and cleaning processes, the next stage is data transformation. This stage aims to convert data that was initially operational in nature into a data structure that is suitable for analysis and reporting needs in the data warehouse. The entire transformation process is carried out using PDI.

The first transformation is carried out to create the dimension table that stores employee descriptive data. At this stage, for each dimension, relevant attributes are selected, a surrogate key is created and lookup, insert, and update mechanism are used according to the needs of each dimension. This aims to ensure that the data integrity of each table is maintained. An overview of the dimension table transformation in Pentaho can be seen in Figure 7.

Apart from transformations on the dimension tables, transformations are also carried out to create a fact table as a repository to store measurements and metrics to be analyzed. The fact table combines various information from the dimension tables through surrogate keys so that relationships between data can be connected to each other efficiently. An illustration of the fact table transformation in Pentaho can be seen in Figure 8.

At this transformation stage, the Slowly Changing Dimension (SCD) mechanism is also applied to the dimension tables according to the need for storing historical data. This is done because, in the real world, employee data often changes; to ensure that tracking can still be performed, the SCD type in each dimension table needs to be adjusted so that both old and new versions of employee data can be traced. This is also very important for trend analysis and the evaluation of employee conditions over time.

With this transformation process, data that was previously stored in an operational format is successfully converted into a more organized, integrated data warehouse structure so that it is ready to be used for analytical needs at the next stage.

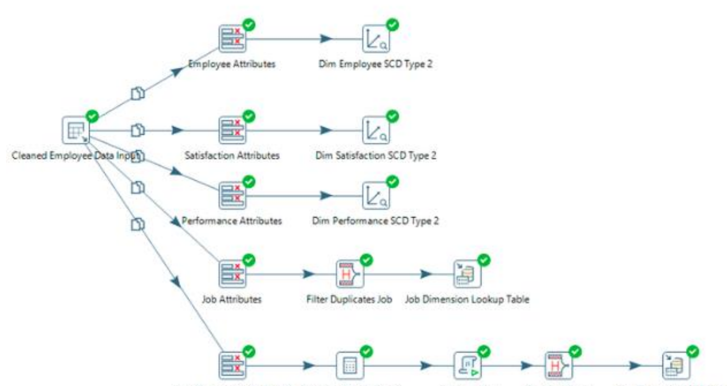

Fig. 7. Dimension Transformation

Fig. 8. Dimension Fact

D. Data Modelling

After the transformation process is complete, the data is loaded into a data warehouse designed using a star schema. This data modeling aims to optimize the analysis and reporting process because the data structure is simple, integrated, and easy to access by Business Intelligence applications. The star schema design used in this study can be seen in Figure 9.

In this study, there is a fact table (Fact_Attrition) which functions as a center for storing the main metrics used in the employee attrition analysis. This table stores various numeric attributes that can be calculated, such as Age, Attrition, Daily Rate, Distance From Home, Hourly Rate, Monthly Income, Monthly Rate, Num Companies Worked, Percent Salary Hike, Total Working Years, Training Times Last Year, Years At Company, Years In Current Role, Years Since Last Promotion, and Years With Curr Manager.

Apart from that, there are also Fact Keys as the Primary Key, as well as Employee Key, Performance Key, Job Key, Satisfaction Key, and Date Key as surrogate keys which are stored as Foreign Keys in the fact table, thereby enabling the multidimensional analysis process to be carried out efficiently. This fact table performs a lookup using Employee Number and also Load Date because the concept of this fact table is full historical.

To help analyze employee attrition in more detail, employee descriptive information is separated into different dimension tables. This is done to reduce data redundancy and also increase flexibility during the reporting process. The dimension tables used in this research consist of:

- Dim_Employee, this dimension stores employee profile information with a surrogate key which acts as the primary key, alongside Employee Number, Gender, Marital Status, Education, Education Field, Business Travel, Overtime, Version, Date From and Date To. This dimension applies SCD Type 2 so that changes to historical data can be saved without deleting previous data.

- Dim_Satisfaction, this dimension stores information related to the level of employee satisfaction with a surrogate key which acts as the primary key, alongside Employee Number, Environment Satisfaction, Job Satisfaction, Relationship Satisfaction, Work Life Balance, Job Involvement, Version, Date From and Date To. This dimension applies SCD Type 2 so that changes in employee satisfaction levels can be saved without deleting previous data.

- Dim_Performance, this dimension stores information related to employee performance with a surrogate key which acts as the primary key, alongside Employee Number, Performance Rating, Stock Option Level, Version, Date From, and Date To. This dimension applies SCD Type 2 so that changes in employee performance levels can be saved without deleting previous data.

- Dim_Job: This dimension stores employee job information, with a surrogate key acting as the primary key, alongside Department, Job Role, and Job Level. This dimension applies SCD Type 0 because job attributes do not change and remain constant during the analysis period.

- Dim_Date, this dimension stores time-related information, with a surrogate key acting as the primary key, alongside Load Date, Load Day, Load Month, and Load Year. This dimension stores information generated from load_date during the ingestion process. This dimension applies SCD Type 0 because date data is static.

The fact table contains data using the Historical Snapshot Fact Table approach. Each integration process automatically produces a new snapshot based on the load date stored in Dim_Date. This helps companies to store all historical data so that changes in employee conditions can be analyzed over time.

Implementing a star schema also helps reduce query complexity, improves analysis performance in Power BI, and simplifies the data aggregation and visualization process. In this way, the dashboard can perform filtering, slicing, and drill-down more quickly than when the data is stored in the form of operational tables.

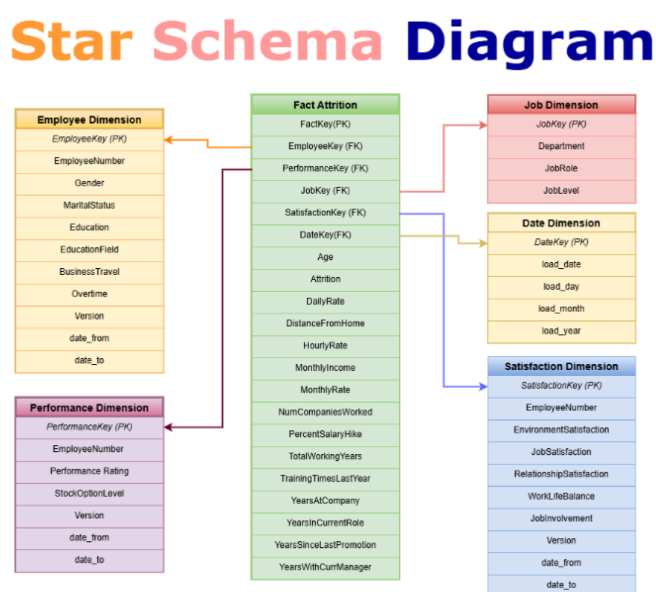

Fig. 9. Star Schema Dimensional Modelling

IV. Data Quality And Validation

Data quality is perhaps the most crucial aspect since it would determine whether the resulting data is credible. This can be done through validation within the entire data pipeline such that the data that goes into analysis meets the required standards.

As for validation, one of the validation processes performed in this research is managing missing values. Missing values will be processed based on predetermined criteria so as not to disturb the analysis process. Meanwhile, data that do not meet the criteria and contain invalid attributes will not proceed to the next process so that the analysis is not affected.

Moreover, all attributes are designed in accordance with the set of rules by performing appropriate consistency checks on the data. The consistency checks are performed through validation testing of the appropriateness of data types, values, and relationships among related attributes. Thus, high-quality data is assured before loading into the data warehouse.

This pipeline also has error logging to record errors. Data that does not successfully pass the validation process is not immediately deleted, but is first instead isolated into specific tables such as "False Positives" and "Bad HR Data Log". This facilitates future investigations if there is a need to examine anomalies in the data in the future.

By implementing various validations, the data used in the analysis stage has good consistency, accuracy, and durability. That way, the information produced by dashboards and reporting can also be used as a basis for more reliable decision-making.

V. Automation And Orchestration

To ensure that the entire process can run smoothly and reduce manual work, this pipeline is automated using Pentaho Jobs. Pentaho Jobs functions as the main orchestrator that manages the sequence of each pipeline stage, starting from ingestion, transformation, and loading data into the data warehouse, to recording system logs. An overview of the orchestration workflow in this study can be seen in Figure 10.

First, execution begins by checking database connectivity to ensure that all storage environments that will be used both OLTP and OLAP, can be accessed properly. This is important to prevent the pipeline from failing due to connection problems and ultimately incomplete data loading. After the connection is successfully verified, the system will continue with staging area and cleaning data process before continuing transformation and loading into the data warehouse.

This orchestration mechanism uses a dependency-based execution approach, which means that each process can only be executed if the previous process has been successfully completed. In this way, data flow can be maintained because if there is an error at one stage, it will not interfere with the next stage. If a failure occurs during the staging, cleaning, transformation, or data loading process, the workflow will automatically be stopped using an abort process mechanism so that invalid data does not enter the analytical environment.

A Pentaho Job is also used to facilitate system monitoring and auditing. Every time a pipeline is run, information such as execution time, process status, and details of stages that were successful or failed to run will be generated. This information will be recorded in the job_log table so that if an investigation or performance evaluation of the pipeline is needed, the pipeline execution history can be traced back.

By implementing this automation and orchestration, the entire data processing process can run more easily, efficiently, consistently, and with more control. This also reduces the risk of human error which often occurs when data processing is done manually. Apart from that, the workflow that has been packaged into one Pentaho Job also allows the pipeline to be integrated with scheduling so that the process can be carried out according to the desired time without having to be carried out directly by humans.

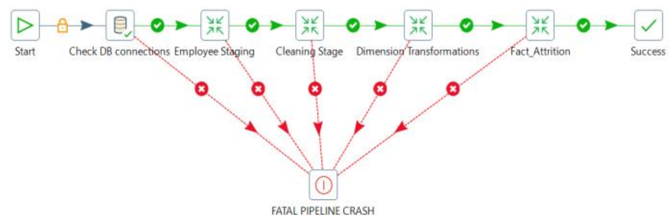

Fig. 10. Job Visualization

VI. Data Consumption (Power BI Dashboard)

The data consumption stage is the last part of this pipeline, which functions to present information to users. In this study, data that has been stored in a star schema-based data warehouse is loaded into Power BI to create an interactive dashboard. This dashboard was created to help the human resources management team understand the company's employee attrition patterns from various analytical perspectives, starting from operational aspects, demographics, and performance, to employee satisfaction levels.

The dashboard developed contains four main pages that complement each other. Each page can answer different analytical needs according to the user's requirements so that the management team can get a more detailed picture of the human resources conditions in the company.

The first page is Executive Attrition & Cost Analytics, which can be seen in Figure 11. This dashboard focuses on the main indicators related to employee attrition and its impact on the company. The metrics displayed consist of the total number of employees working in the company, the number of employees experiencing attrition, and the overall attrition percentage rate. Apart from that, the visualizations also show the relationship between the number of attritions and salary allocations based on position, the distribution of the payroll budget in each department, as well as attrition patterns based on employee tenure. From the dashboard, it can be seen that most of the attrition occurs among employees with a fairly short work period, namely 0–1 year.

The second page is Workforce Demographics & Profile, which can be seen in Figure 12. This dashboard focuses on the demographic characteristics of employees, such as age distribution, gender, marital status and also the employee educational background. The metrics displayed consist of the average age of employees, the gender ratio of the workforce, and the distribution of employee marital status. Apart from that, the visualizations also helps show employee groups that have higher levels of attrition than other groups. From this dashboard it can be seen that the largest attrition ratio comes from employees with a Human Resources educational background.

The third page is Performance & Talent Retention, which can be seen in Figure 13. This dashboard focuses on employee performance and the relationship between work experience and employee performance levels. The metrics displayed consist of the average employee performance, average length of service, as well as the relationship between total work experience and performance in each department. This dashboard can be used to create talent development and employee retention strategies.

The fourth page is Employee Satisfaction & Engagement, which can be seen in Figure 14. This dashboard focuses on the level of employee satisfaction and engagement, which is often an important factor in attrition. The metrics displayed consist of job satisfaction levels, work-life balance, and comparisons between positions. From this dashboard, it can be seen that certain positions have a higher attrition rate than other positions. This should be a key consideration when developing employee retention strategies.

With the integration of the data warehouse and Power BI, the analysis process can be carried out more quickly and interactively. The resulting dashboard not only helps the human resources management team to understand the current condition of employees but also helps identify patterns and factors that can influence employee attrition within the company.

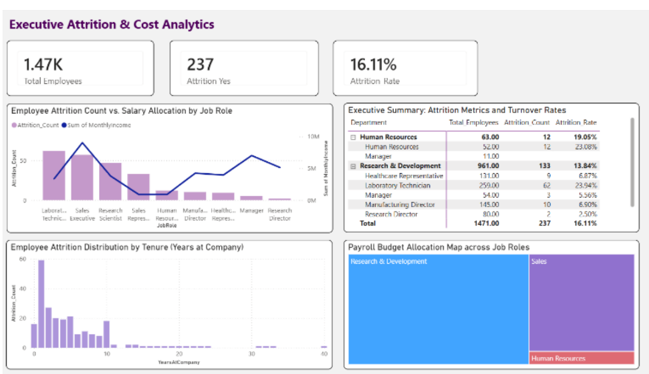

Fig. 11. Executive Attrition & Cost Analytics

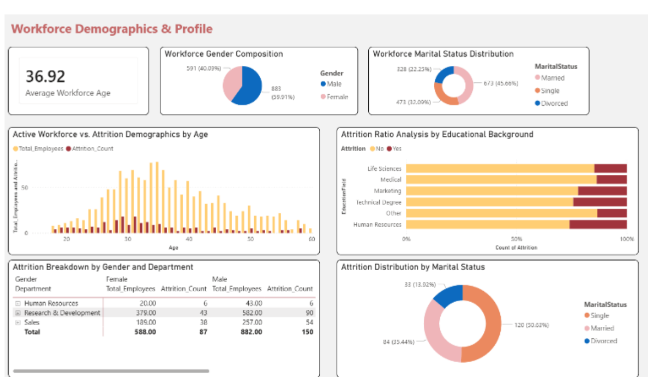

Fig. 12. Workforce Demographics & Profile

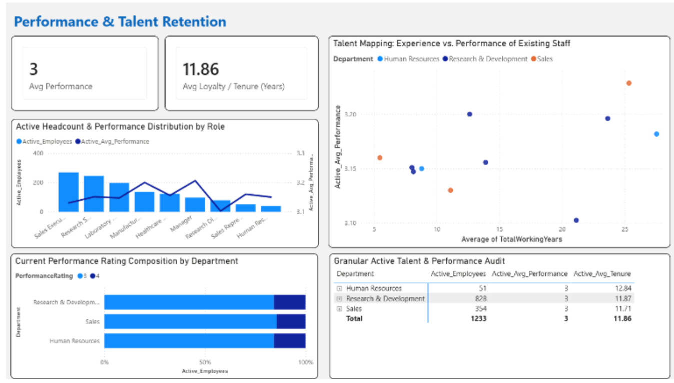

Fig. 13. Performance & Talent Retention

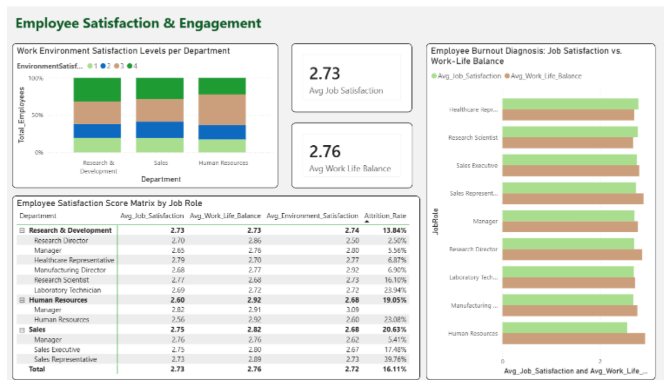

Fig. 14. Employee Satisfaction & Engagement

VII. Evaluation And Reflection

The pipeline in this study uses several technologies selected based on data processing needs, ease of use, and the ability to integrate system components. Evaluation is carried out on the pipeline that has been created to assess the effectiveness of the architecture used to support employee attrition analysis.

Pentaho Data Integration (PDI) is used as a platform for the ETL process because it has an easy-to-learn UI that makes it easier to design workflows, data transformations, and automated pipelines. Apart from that, PDI also has the features needed for lookups, SCD, and job orchestration so that the process can be carried out in a more structured and easy-to-maintain manner.

For data storage, MySQL is used as the database for both OLTP and OLAP. The two databases are separated so that the data flow is more organized and data management and monitoring in the pipeline are easier. The data warehouse structure built using the star schema approach also supports multidimensional analysis needs more efficiently.

Power BI is used for visualization because it is able to connect to warehouse data directly; apart from that, Power BI also has many interactive analytical features. With a structured data model, the dashboard creation process becomes simpler and produces optimal query performance when exploring data.

EDA is carried out using Python in the early stages of the research. This facilitates easier identification of data characteristics within the dataset, anomaly and outlier detection, and an initial understanding of the attributes to be utilized.

Despite the success achieved in meeting the analytical requirements of the dataset through this implementation, certain limitations must be considered. While loading the entire fact table is efficient for small- to medium-sized datasets, its efficiency decreases as data volume increases. Consequently, future developments should consider incorporating incremental loads or real-time data processing.

VIII. Conclusion

The study was able to successfully create an end-to-end data engineering pipeline that uses tools such as Pentaho Data Integration, MySQL, and Power BI in analyzing attrition of employees. The pipeline covers all steps of data engineering ranging from data ingestion and storage to transformation, modeling, and visualization. Using the star schema in the OLAP database makes the data more effective for analysis.

The implementation results also show that the pipeline created was successful in automating the data processing with maintained data quality and consistency, because the various validation mechanisms were implemented. The processed data is then used to create a Power BI dashboard that displays information about attrition patterns based on employee characteristics, work conditions, performance and job satisfaction levels.

In general, this study proves that through using this pipeline, it will be useful to convert raw data into insights and facilitating data-driven decisions in the field of human resource management. Through the use of integrated and accessible information, companies will be able to analyze the factors of employee attrition.

Going forward, there is room for this study to be extended further through the development of a machine learning system capable of predicting the probability of attrition in the future. In doing so, this system will not only be a platform for analysis but prediction too.

IX. Open Data

The dataset used in this research is the HR Analytics public dataset, which can be obtained via the Kaggle platform and accessed at: Link Dataset

All materials used and produced from this study—including transformation files, EDA, SQL files, dashboards, presentation materials, the paper, design diagrams, jobs, logs, and datasets—can be accessed from the following GitHub repository: https://github.com/Briant-s/DE_Group2

By providing this dataset and source code, this study aims to support research transparency and enable further studies related to employee attrition analysis and pipeline implementation.

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